SPINORS OF EXCEPTIONAL SUPER YANG–MILLS THEORIES ARE NOT BRANE COHOMOLOGIES

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Abstract. Rios, Marrani and Chester identified the Majorana–Weyl spinors of their exceptional super Yang–Mills theories in $(8n+3)+3$ dimensions with the total cohomology of the largest brane. We prove that no such identification exists for any $n \geq 1$: on every spinor the Casimir operator of the brane rotation algebra takes one value that no exterior power attains. Their dual M-theory in $13+1$ dimensions does not arise from the reduction they describe either.

1. Introduction

Rios, Marrani and Chester proposed in [10] a family of supersymmetric Yang–Mills theories in the signatures $(9+8n)+1$, $(10+8n)+2$, $(11+8n)+3$ and $(12+8n)+4$, which they called exceptional, by moving $4n$ timelike directions of Nishino's signatures [9] into space; supersymmetry in these dimensions goes back to [9, 12]. In [11] they took up the case $27+3$, in which the supersymmetry algebra

$$\{Q_\alpha, Q_\beta\} = (\gamma^{\mu\nu\rho})_{\alpha\beta} Z_{\mu\nu\rho} + (\gamma^{\mu_1\cdots\mu_7})_{\alpha\beta} Z_{\mu_1\cdots\mu_7} + (\gamma^{\mu_1\cdots\mu_{11}})_{\alpha\beta} Z_{\mu_1\cdots\mu_{11}} + (\gamma^{\mu_1\cdots\mu_{15}})_{\alpha\beta} Z_{\mu_1\cdots\mu_{15}} \tag{1}$$

carries the charges of a 3-, 7-, 11- and 15-brane. That these theories cannot be built by repeating Nishino's construction is shown in [2]; related geometric questions for Yang–Mills fields and for compactifications are treated in [4, 3]. The present paper is about a different claim of [11], made in its Section III and intended to extend the observation of Dijkgraaf, Verlinde and Verlinde [5] that the sixteen central charges of the five-brane are fluxes through its homology cycles.

The claim reads as follows. Write $\Lambda^k\mathbf{15}$ for the k-th exterior power of the vector representation of $\mathfrak{so}_{15}$. Equations (3.4) and (3.5) of [11] state that

$$\underbrace{\mathbf{16384}}_{\mathfrak{so}_{27,3}\text{ MW spinor}} = \bigoplus_i \Lambda^{2i+1}\mathbf{15}, \qquad \underbrace{\mathbf{16384'}}_{\mathfrak{so}_{27,3}\text{ MW conj. spinor}} = \bigoplus_i \Lambda^{2i}\mathbf{15}, \tag{2}$$

covariantly under $\mathfrak{so}_{15}$, and call the right-hand sides the odd and even cohomology of the 15-brane, the largest brane in (1). The text before (3.4) says that the semispinor **32768** of $\mathfrak{so}_{28,4}$ is the total cohomology of the same brane, and (3.6)–(3.8) assert, as "a general fact", that for every level n the semispinor $\mathbf{2}^{4n+3}$ of $\mathfrak{so}_{8n+4,4}$ is the total cohomology of a $(4n+3)$-brane; for $n=1$ the **128** of $\mathfrak{e}_{8(-24)}$ is said to be the total cohomology of the 7-brane.[1]

Dimension counts cannot decide such a statement, since many modules share a dimension. The question is whether some subalgebra of $\mathfrak{so}_{s,t}$ isomorphic to $\mathfrak{so}_{4n+3}$ makes the spinor module isomorphic to a sum of exterior powers of its vector module. For the five-brane of [5] it does: there $\mathfrak{so}_5 \subset \mathfrak{gl}_5 \subset \mathfrak{so}_{5,5}$ acts on $\mathbb{R}^{5,5} = \mathbb{R}^5 \oplus (\mathbb{R}^5)^*$, and the spinors of $\mathfrak{so}_{5,5}$ restrict to $\Lambda^\bullet\mathbb{R}^5$ up to a power of the determinant [6, §20.2]. We show that in the signatures of [10, 11] it never does.

Theorem A. *Let $n \geq 2$ and $m = 4n+3$.*

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[1] The dimensions agree: $\sum_k \binom{15}{k} = 2^{15}$ and $\sum_{k\ \text{odd}} \binom{15}{k} = 2^{14}$. The list printed in (3.5) of [11] ends in $\mathbf{25'}$, which should be $\binom{15}{14} = 15$; with 25 the list adds up to 16394.

(a) *Let $\mathfrak{h} \subset \mathfrak{so}_{8n+3,3}$ be a subalgebra isomorphic to a real form of $\mathfrak{so}_m(\mathbb{C})$. No exterior power of the vector module of $\mathfrak{h}$ occurs in either semispinor of $\mathfrak{so}_{8n+3,3}$, even after complexification.*

(b) *Let $\mathfrak{h} \subset \mathfrak{so}_{8n+4,4}$ be a subalgebra isomorphic to the compact algebra $\mathfrak{so}_m$. No exterior power of the vector module of $\mathfrak{h}$ occurs in either semispinor of $\mathfrak{so}_{8n+4,4}$, even after complexification.*

For $n = 3$ this says that (2) is false for every choice of $\mathfrak{so}_{15}$ inside $\mathfrak{so}_{27,3}$, and that the **32768** of $\mathfrak{so}_{28,4}$ is not the cohomology of a 15-brane; for $n = 2$ it says the same about $19+3$ and $20+4$. The case $n = 1$ needs a separate argument, because $\mathfrak{so}_7$ has an eight-dimensional spin module that fits into $\mathbb{R}^{12,4}$.

Theorem B. *Let $\mathfrak{h}$ be a subalgebra of $\mathfrak{so}_{11,3}$ or of $\mathfrak{so}_{12,4}$ isomorphic to the compact algebra $\mathfrak{so}_7$. Neither semispinor contains $\Lambda^2\mathbb{R}^7$, $\Lambda^3\mathbb{R}^7$, $\Lambda^4\mathbb{R}^7$ or $\Lambda^5\mathbb{R}^7$, even after complexification. In particular the **128** of $\mathfrak{so}_{12,4}$ is not $\Lambda^\bullet\mathbb{R}^7$ and the **64** of $\mathfrak{so}_{11,3}$ is not $\bigoplus_i \Lambda^{2i+1}\mathbb{R}^7$.*

The obstruction is the signature. Over $\mathbb{C}$ the identification holds, through $\mathfrak{so}_m(\mathbb{C}) \subset \mathfrak{gl}_m(\mathbb{C}) \subset \mathfrak{so}_{2m}(\mathbb{C})$, and Remark 5.1 shows that it also holds for the noncompact real form $\mathfrak{so}_{4n+1,2}$ inside $\mathfrak{so}_{8n+4,4}$; but the rotations of the spatial worldvolume of a brane form a compact algebra, and for $\mathfrak{so}_{8n+3,3}$ no real form helps. The proofs rest on one computation, Lemma 2.1: on every spinor the Casimir operator of $\mathfrak{so}_m$ equals $m(m-1)/8$, which is not an integer when $m \equiv 3 \pmod 4$, while on Λ^k it equals the integer $k(m-k)$ (Figure 1).

Section II of [11] also introduces a "dual M-theory" in $13+1$ dimensions as the worldvolume theory of the magnetic dual of the 10-brane, obtained by reduction from $26+2$ to $26+1$.

Proposition C. *In the conventions of* [11]*, the reduction $26+2 \to 26+1$ takes the 14-brane with worldvolume $14+2$ to a 14-brane with worldvolume $14+1$, not to the 13-brane with worldvolume $13+1$; and the 10-brane and the 14-brane are not electric–magnetic duals in $26+1$ dimensions.*

Section 2 fixes notation, Section 3 bounds the small modules of $\mathfrak{so}_{2l+1}$, Section 4 shows how such an algebra can sit in $\mathfrak{so}_{s,t}$, Section 5 proves Theorems A and B, and Section 6 proves Proposition C. The finite arithmetic used below is rechecked by a short program in the archive of this paper [1]; no proof depends on it.

2. Preliminaries

Let $V = \mathbb{R}^{s,t}$ with metric η of signature (s,t), and $V_{\mathbb{C}} = V \otimes \mathbb{C}$. The Clifford relations are $vw + wv = 2\eta(v,w)$. When $s+t = 2r$ the complex Clifford algebra $\mathrm{Cl}(V_{\mathbb{C}})$ has a unique irreducible module S, of dimension 2^r, and $S = S^+ \oplus S^-$ under the chirality operator; $S^\pm$ are the semispinors. The Lie algebra $\mathfrak{so}(V_{\mathbb{C}})$ acts on S through

$$E_{v,w} \longmapsto \tfrac{1}{2}\, vw \qquad (\eta(v,w) = 0), \tag{3}$$

where $E_{v,w}(x) = \eta(w,x)\, v - \eta(v,x)\, w$; indeed $\frac{1}{2}(vwx - xvw) = \eta(w,x)v - \eta(v,x)w$ for $\eta(v,w) = 0$ [8, Ch. I, §6]. A Majorana–Weyl spinor of $\mathfrak{so}_{s,t}$ is a real form of S^+ or S^-; it exists when $s - t \equiv 0 \pmod 8$, as in all signatures considered here. A real $\mathfrak{so}_{s,t}$-module and its complexification have the same submodule structure under any subalgebra up to complexification, so it suffices to prove Theorems A and B for $S^\pm$.

Let $U \subset V_{\mathbb{C}}$ be a nondegenerate subspace of dimension m with a basis $u_1, \dots, u_m$ satisfying $\eta(u_i, u_j) = \delta_{ij}$. Write $E_{ij} = E_{u_i,u_j}$; these span $\mathfrak{so}(U)$, which acts on $V_{\mathbb{C}}$ and is zero on $U^\perp$. The Casimir element

$$\Omega = -\sum_{1 \le i < j \le m} E_{ij}^2 \tag{4}$$

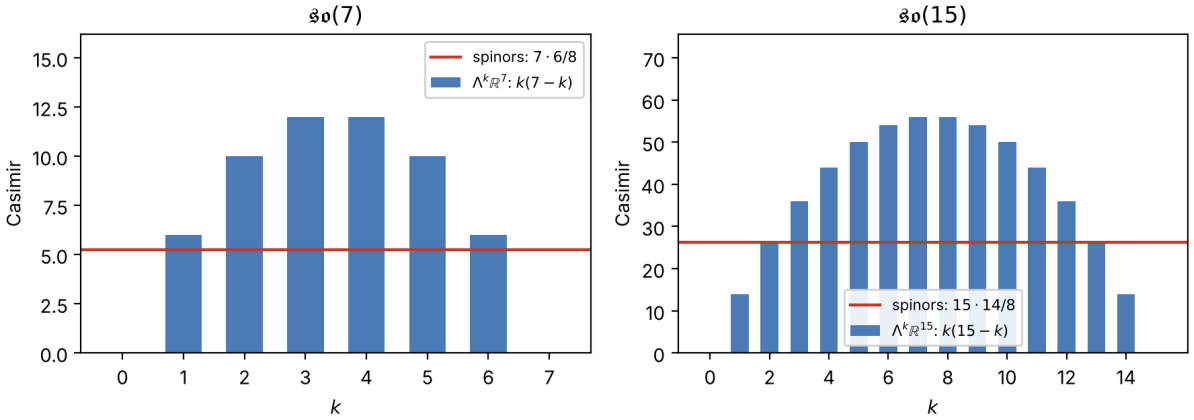


FIGURE 1. The Casimir operator (4) on the exterior powers $\Lambda^k\mathbb{R}^m$ (bars) and on spinors (line), for $m = 7$ and $m = 15$. The line meets no bar.

is central in the enveloping algebra of $\mathfrak{so}(U)$, so it acts by a scalar on every irreducible $\mathfrak{so}(U)$-module and every module map between two modules intertwines it.

Lemma 2.1. (i) *On $\Lambda^k U$ the element Ω acts by $k(m-k)$.*
(ii) *On S, and hence on S^+ and S^-, the element Ω acts by $m(m-1)/8$.*
(iii) *If $m \equiv 3 \pmod 4$, then $m(m-1)/8 \neq k(m-k)$ for every integer k.*

Proof. (i) We have $E_{ij}u_j = u_i$, $E_{ij}u_i = -u_j$ and $E_{ij}u_l = 0$ for $l \notin \{i,j\}$, and E_{ij} acts on $\Lambda^k U$ as a derivation. Let $u_I = u_{i_1} \wedge \cdots \wedge u_{i_k}$ with $I = \{i_1 < \cdots < i_k\}$. If neither i nor j lies in I, then $E_{ij}u_I = 0$. If both lie in I, then $E_{ij}u_I$ is a sum of two wedge products each containing a repeated vector, so it vanishes. If exactly one lies in I, then E_{ij} exchanges u_i and u_j up to sign and $E_{ij}^2 u_I = -u_I$. Exactly $k(m-k)$ pairs $\{i,j\}$ have one element in I, so $\Omega u_I = k(m-k)u_I$.

(ii) By (3), E_{ij} acts on S as $\frac{1}{2}u_iu_j$, and for $i \neq j$

$$\left(\tfrac{1}{2}u_iu_j\right)^2 = \tfrac{1}{4}u_iu_ju_iu_j = -\tfrac{1}{4}u_i^2u_j^2 = -\tfrac{1}{4}.$$

Summing over the $\binom{m}{2}$ pairs gives $\Omega = \frac{1}{4}\binom{m}{2} = m(m-1)/8$ on all of S.

(iii) If $m = 4n+3$, then $m(m-1)/8 = (4n+3)(2n+1)/4$, whose numerator is odd, so it is not an integer, while $k(m-k)$ is. □

3. SMALL MODULES OF $\mathfrak{so}_{2l+1}$

Lemma 3.1. *Let $l \geq 3$ and let $W = \mathbb{C}^{2l+1}$ be the vector module of $\mathfrak{so}_{2l+1}(\mathbb{C})$. Every irreducible $\mathfrak{so}_{2l+1}(\mathbb{C})$-module of dimension less than $l(2l+1)$ is trivial, isomorphic to W, or isomorphic to the spin module, of dimension 2^l.*

Proof. Let $\omega_1, \ldots, \omega_l$ be the fundamental weights, so that the irreducible module $V(\omega_i)$ is $\Lambda^i W$ for $i < l$ and $V(\omega_l)$ is the spin module [6, §19.5]. By the Weyl dimension formula,

$$\dim V(\lambda) = \prod_{\alpha>0} \frac{(\lambda+\rho,\alpha)}{(\rho,\alpha)},$$

and for dominant weights λ, μ each factor for $\lambda+\mu$ is at least the corresponding factor for λ, so $\dim V(\lambda+\mu) \geq \dim V(\lambda)$. Let $\lambda = \sum_i a_i\omega_i$ be dominant.

If $a_i \geq 1$ for some $2 \leq i \leq l-1$, then $\dim V(\lambda) \geq \binom{2l+1}{i} \geq \binom{2l+1}{2} = l(2l+1)$, because the binomial coefficients increase up to $i = l$. Otherwise $\lambda = a\omega_1 + b\omega_l$. If $a \geq 2$, then $\dim V(\lambda) \geq \dim V(2\omega_1)$, and $V(2\omega_1)$ is the traceless part of $\mathrm{Sym}^2 W$, of dimension $(2l+1)(l+1) - 1 = l(2l+3) > l(2l+1)$. If $b \geq 2$, then $\dim V(\lambda) \geq \dim V(2\omega_l) = \dim \Lambda^l W = \binom{2l+1}{l} \geq l(2l+1)$. If $a \geq 1$ and $b \geq 1$, then $\dim V(\lambda) \geq \dim V(\omega_1 + \omega_l)$, and $W \otimes V(\omega_l) = V(\omega_1 + \omega_l) \oplus V(\omega_l)$ gives $\dim V(\omega_1 + \omega_l) = 2l \cdot 2^l \geq l(2l+1)$. The remaining weights are 0, ω_1 and ω_l. □

Corollary 3.2. *Let $m = 4n+3$. If $n \geq 2$, every $\mathfrak{so}_m(\mathbb{C})$-module of dimension at most $8n+8$ is a sum of trivial modules and copies of W. If $n = 1$, every $\mathfrak{so}_7(\mathbb{C})$-module of dimension at most 16 is a sum of trivial modules, copies of $W = \mathbb{C}^7$ and copies of the spin module $\mathbb{C}^8$.*

Proof. Here $l = 2n+1$, and $8n+8 < l(2l+1) = (2n+1)(4n+3)$ for all $n \geq 1$. For $n \geq 2$ also $8n+8 < 2^{2n+1}$: this holds for $n = 2$ ($24 < 32$), and passing from n to $n+1$ adds 8 on the left and multiplies the right by 4. For $n = 1$, $16 < 21$. Finite-dimensional modules of $\mathfrak{so}_m(\mathbb{C})$ are completely reducible. □

4. Embeddings of $\mathfrak{so}_m$ in $\mathfrak{so}_{s,t}$

Lemma 4.1. *Let $\mathfrak{h} \subset \mathfrak{so}_{s,t}$ be a subalgebra isomorphic to a real form of $\mathfrak{so}_m(\mathbb{C})$, with m odd, and let $V_0 = \{v \in V : \mathfrak{h}v = 0\}$ and $V_1 = \mathfrak{h}V$. Then $V = V_1 \oplus V_0$, the two summands are orthogonal and η is nondegenerate on each.*

(a) *If $(s,t) = (8n+3, 3)$, $m = 4n+3$ and $n \geq 2$, then $V_1 \otimes \mathbb{C}$ is isomorphic to the vector module W of $\mathfrak{h} \otimes \mathbb{C}$.*
(b) *If $(s,t) = (8n+4, 4)$, $m = 4n+3$, $n \geq 2$ and $\mathfrak{h}$ is compact, then $V_1 \otimes \mathbb{C} \cong W$ and V_1 is positive definite.*
(c) *If $(s,t) \in \{(11,3), (12,4)\}$ and $\mathfrak{h} \cong \mathfrak{so}_7$ is compact, then V_1 is positive definite of dimension 7 or 8, and $\mathfrak{h}$ acts on it by the vector or the spin module.*

Proof. The module V is completely reducible and V_0 is its isotypic component of trivial type, so $V = V_1 \oplus V_0$. For $x \in V_0$, $X \in \mathfrak{h}$ and $z \in V$ we have $\eta(x, Xz) = -\eta(Xx, z) = 0$, so $V_0 \perp V_1$, and since η is nondegenerate on V it is nondegenerate on both summands. The algebra $\mathfrak{so}_{s,t}$ acts faithfully on V, hence so does $\mathfrak{h}$, and $V_1 \neq 0$.

Every real form of $\mathfrak{so}_m(\mathbb{C})$ with m odd is some $\mathfrak{so}_{p,q}$ with $p+q = m$, and its vector module $\mathbb{R}^{p,q}$ is absolutely irreducible and carries an invariant form b of signature (p,q), unique up to a factor. If $V_1 \otimes \mathbb{C} \cong W \oplus W$, then $V_1 \cong \mathbb{R}^{p,q} \otimes \mathbb{R}^2$, and every invariant symmetric form on it is $b \otimes M$ with M a symmetric 2×2 matrix, nondegenerate here. Diagonalising M shows that the signature of V_1 is $(2p, 2q)$, (m, m) or $(2q, 2p)$.

(a) By Corollary 3.2, $V_1 \otimes \mathbb{C}$ is W or $W \oplus W$. In the second case $V_1 = V$ has dimension $2m = 8n+6$ and signature $(8n+3, 3)$; but $8n+3$ is odd, and $m > 3$, so none of the three signatures above is possible.

(b) Now $p = m$, $q = 0$. If $V_1 \otimes \mathbb{C} \cong W \oplus W$, then V_1 has signature $(2m, 0)$ or (m, m) and V_0 has dimension 2, so the number of negative directions of V is at most 2 or at least $m > 4$, a contradiction. Hence $V_1 \otimes \mathbb{C} \cong W$, and b is definite; since V_1 has dimension $m > 4$, it is positive definite.

(c) By Corollary 3.2, V_1 is a sum of copies of $\mathbb{R}^7$ and of the spin module $\mathbb{R}^8$, both carrying definite invariant forms unique up to a factor. The isotypic components are orthogonal, and as above an isotypic component with c copies of a module of dimension d has signature $(c'd, c''d)$ with $c' + c'' = c$. With $d \in \{7, 8\}$ and total dimension 14 or 16, the decompositions with at least two nontrivial summands are $\mathbb{R}^7 \oplus \mathbb{R}^7$, $\mathbb{R}^7 \oplus \mathbb{R}^8 \oplus \mathbb{R}$ and $\mathbb{R}^8 \oplus \mathbb{R}^8$, up to trivial summands; the numbers of negative directions of V_1 then lie in $\{0, 7, 14\}$, $\{0, 7, 8, 15\}$ and $\{0, 8, 16\}$. Since V_0

has dimension at most 2, the space V would have at most 2 or at least 7 negative directions, never 3 or 4. So V_1 is one copy of $\mathbb{R}^7$ or of $\mathbb{R}^8$, definite, hence positive definite because $t < 7$. □

5. Proofs of Theorems A and B

Proof of Theorem A. By Lemma 4.1(a) or (b), $U = V_1 \otimes \mathbb{C}$ is a nondegenerate subspace of $V_{\mathbb{C}}$ of dimension m on which $\mathfrak{h} \otimes \mathbb{C}$ acts by its vector module, and $\mathfrak{h}$ is zero on $U^\perp = V_0 \otimes \mathbb{C}$. Since $\mathfrak{h} \otimes \mathbb{C}$ acts faithfully on U and preserves η, and $\dim \mathfrak{h} = \dim \mathfrak{so}(U)$, the image of $\mathfrak{h} \otimes \mathbb{C}$ in $\mathfrak{so}(V_{\mathbb{C}})$ is $\mathfrak{so}(U)$, and the vector module of $\mathfrak{h}$ is U. By Lemma 2.1(ii) the Casimir element of $\mathfrak{so}(U)$ acts on S^+ and S^- by $m(m-1)/8$, and by Lemma 2.1(i) it acts on $\Lambda^k U$ by $k(m-k)$. These numbers differ by Lemma 2.1(iii), so there is no nonzero $\mathfrak{h}$-module map $\Lambda^k U \to S^\pm$. □

In the language of [11]: under any $\mathfrak{so}_{15}$ inside $\mathfrak{so}_{27,3}$ the module **16384** is 128 copies of the spin module **128** of $\mathfrak{so}_{15}$, and it contains none of the modules **15**, **455**, **3003**, **6435**, **5005**, **1365**, **105**, **1** of (2). Indeed, choose a nondegenerate line $\ell \subset U^\perp$ and put $U' = U \oplus \ell$. Then $S \cong S_{U'} \otimes S_{U'^\perp}$ as a module over $\mathfrak{so}(U') \oplus \mathfrak{so}(U'^\perp)$, and both semispinors of $\mathfrak{so}(U') \cong \mathfrak{so}_{16}(\mathbb{C})$ restrict to the spin module of $\mathfrak{so}(U) \cong \mathfrak{so}_{15}(\mathbb{C})$, so $\mathfrak{so}(U)$ acts on S through copies of its spin module.

Proof of Theorem B. By Lemma 4.1(c) there are two cases.

If $V_1 \cong \mathbb{R}^7$, the argument of Theorem A applies with $m = 7$: the Casimir element acts by $21/4$ on $S^\pm$ and by an integer on every $\Lambda^k \mathbb{R}^7$.

If $V_1 \cong \mathbb{R}^8$ with $\mathfrak{h}$ acting by the spin module, let $U = V_1 \otimes \mathbb{C}$. Then $\mathfrak{h} \otimes \mathbb{C}$ lies in $\mathfrak{so}(U) \cong \mathfrak{so}_8(\mathbb{C})$ as the image of $\mathfrak{so}_7(\mathbb{C})$ under its spin representation. The three eight-dimensional modules of $\mathfrak{so}_8(\mathbb{C})$, the vector module U and the semispinors S^+_U, S^-_U, are permuted by triality, and the standard $\mathfrak{so}_7$, the stabiliser of a vector, acts on them by $\mathbb{C}^7 \oplus \mathbb{C}$, $\mathbb{C}^8$, $\mathbb{C}^8$. Applying triality, the subalgebra $\mathfrak{h} \otimes \mathbb{C}$ acts on U by $\mathbb{C}^8$ and on $\{S^+_U, S^-_U\}$ by $\mathbb{C}^7 \oplus \mathbb{C}$ and $\mathbb{C}^8$ in some order [6, §20.3]. Since $U^\perp$ has even dimension $2j$ ($j = 3$ or 4),

$$S^+ \cong S^+_U \otimes S^+_{U^\perp} \oplus S^-_U \otimes S^-_{U^\perp}, \qquad S^- \cong S^+_U \otimes S^-_{U^\perp} \oplus S^-_U \otimes S^+_{U^\perp}$$

as modules over $\mathfrak{so}(U) \oplus \mathfrak{so}(U^\perp)$, with $\dim S^\pm_{U^\perp} = 2^{j-1}$. Hence each of S^+ and S^- is, as an $\mathfrak{h}$-module, 2^{j-1} copies of $\mathbb{C} \oplus \mathbb{C}^7 \oplus \mathbb{C}^8$. By Lemma 2.1, applied to $\mathbb{C}^7 \subset \mathbb{C}^8$, whose semispinors restrict to the spin module of $\mathfrak{so}_7(\mathbb{C})$, the Casimir element of $\mathfrak{h}$ acts on these summands by 0, 6 and $21/4$, and on $\Lambda^k \mathbb{C}^7$ by $k(7-k)$, which is 10 for $k = 2, 5$ and 12 for $k = 3, 4$. So none of these four exterior powers occurs in $S^\pm$. Each of $\bigoplus_k \Lambda^k \mathbb{R}^7$, $\bigoplus_i \Lambda^{2i} \mathbb{R}^7$ and $\bigoplus_i \Lambda^{2i+1} \mathbb{R}^7$ contains one of them. □

Remark 5.1. Compactness in Theorem A(b) cannot be dropped. Let $p = 4n + 1$ and $\mathfrak{h} = \mathfrak{so}_{p,2}$ act diagonally on $\mathbb{R}^{p,2} \oplus \mathbb{R}^{p,2}$, and let $V = \mathbb{R}^{p,2} \oplus \mathbb{R}^{p,2} \oplus \mathbb{R}^{2,0}$, of signature $(8n + 4, 4)$. Over $\mathbb{C}$ the space $U = \mathbb{C}^m \oplus \mathbb{C}^m$ contains the isotropic subspaces $L_\pm = \{(u, \pm iu)\}$, each $\mathfrak{h}$-stable and isomorphic to $\mathbb{C}^m$, with $U = L_+ \oplus L_-$; so $\mathfrak{h} \otimes \mathbb{C} \subset \mathfrak{gl}(L_+)$ and the spinors of $\mathfrak{so}(U)$ are $\Lambda^{\mathrm{even}} L_+$ and $\Lambda^{\mathrm{odd}} L_+$, the determinant twist being trivial on $\mathfrak{so}_m$ [6, §20.2]. Tensoring with the two semispinors of the plane $\mathbb{R}^{2,0}$ shows that each semispinor of $\mathfrak{so}_{8n+4,4}$ restricts to $\Lambda^\bullet \mathbb{C}^m$. The algebra $\mathfrak{so}_{p,2}$ is not the rotation algebra of a spatial worldvolume, and Lemma 4.1(a) excludes this construction in $\mathfrak{so}_{8n+3,3}$, where the dimension leaves no room for the extra plane and the signature has odd entries.

Remark 5.2. The argument says nothing against the algebras $\mathfrak{e}^{(n)}_{8(-24)} = \mathfrak{so}_{8n+4,4} \oplus \mathbf{2}^{4n+3}$ of [13] or their grading; it shows only that their spinor generators are not exterior algebras over the rotations of a brane. Other parts of Section III of [11] contain smaller slips: the sums in (3.4) and (3.5) run up to $i = 15$ instead of $i = 7$, and the last term of (3.8) contracts a matrix with $n + 3$ indices against a $(4n + 3)$-form.

6. The dual M-theory

We use the conventions of [11]. In a spacetime of signature $s+t$, a p-brane carries a p-form charge and has a worldvolume of signature $p+t$: the 11-brane of $27+3$ has worldvolume $11+3$, the 10-brane of $26+2$ has worldvolume $10+2$, and the 10-brane of $26+1$ has worldvolume $10+1$. In a spacetime of dimension D with one time, a p-brane couples to a $(p+1)$-form potential with a $(p+2)$-form field strength, whose Hodge dual has degree $D-p-2$; so the magnetic dual of a p-brane is a $(D-p-4)$-brane, as for the 2-brane and the 21-brane of bosonic M-theory in $26+1$ dimensions [7]. The pairs $11 \leftrightarrow 15$ in $27+3$ and $10 \leftrightarrow 14$ in $26+2$ quoted in [11] also satisfy $p+p'=D-4$.

Proof of Proposition C. The reduction $26+2 \to 26+1$ removes one timelike direction and no spacelike one. In the conventions above it takes the 10-brane with worldvolume $10+2$ to the 10-brane with worldvolume $10+1$, as [11] states, and by the same rule it takes the 14-brane with worldvolume $14+2$ to a 14-brane with worldvolume $14+1$. Passing from $14+2$ to $13+1$ would remove a spacelike direction of the worldvolume, which the reduction does not do. Moreover, in $D=27$ the magnetic dual of a 10-brane is a $27-10-4=13$-brane, while $10+14=24 \neq 23$, so the reduced 14-brane is not the dual of the reduced 10-brane. The "dual M-theory" on a $13+1$ worldvolume is therefore not the reduction of the 14-brane of $26+2$. □

The reduction $27+3 \to 26+2$ removes one spacelike and one timelike direction, and there the pairing is preserved: the 11-brane and the 15-brane wrapping the removed spacelike direction become the 10-brane and the 14-brane, and $10+14=28-4$. The pairing fails only at the last step.

Declarations

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